

Faker, Patch Adaptation, and Performance Stability in Professional League of Legends

DSA 210 Final Report

A Faker-centered study of performance stability, champion-pool diversity, and post-patch prediction

Executive Summary

This project aims to explore and compare the ability of a small set of elite mid-lane players to adapt to significant game changes, to that of the most popular professional League of Legends player, Faker. League of Legends is a frequently updated multiplayer game of 5 players per team, and when the game is updated, it is called a patch, and those updates could affect the strength of different characters and, in turn, professional play.

Public match data and champion metadata are joined, with a subsample of 732 match-level observations from Faker, Chovy, ShowMaker, Caps, knight and Bdd, and three key 2024 patch windows are analysed (14.5, 14.13, and 14.16). The project is done in two phases: exploratory analysis and hypothesis testing followed by supervised machine learning on the post-patch sample.

Descriptive results are consistent in the direction of the hypothesis that Faker is a bit more stable than peers after patches, with a mean absolute change in performance of 0.251 vs. 0.291 for peers. He also played fewer new champions immediately after patches (0.228 versus 0.481). The formal hypothesis tests were not statistically significant at conventional levels, however.

The best model for post-patch stability was the random forest (accuracy 0.624, ROC-AUC 0.639) and the best model for predicting high-performance post-patch matches was logistic regression (accuracy 0.681, ROC-AUC 0.704) in the machine learning stage. The findings indicate that patch adaptation is not entirely predictable based on recent performance and context, but generalization to future patches is still difficult.

1. Motivation

The Professional League of Legends constantly evolves as Riot Games continually patches the game to make changes to champions, items, and the broader game. In the highest levels of play, then, being able to adapt is as important as possessing raw talent. This serves as an interesting environment for a data science project on stability, flexibility and prediction under varying conditions in League of Legends.

The interesting aspect of Faker is that he is consistently a top player across many years and many versions of the game. This project does not approach him as an individual celebrity, but instead sees him as being in a small benchmark group of other elite mid-lane players against which the question can be measured and compared.

The main research question is: Does Faker's performance remain stable in comparison to other high-level players in the mid lane after major game changes, and does this relate to the diversity of the champion pool? In this context, the champion-pool diversity is the number of characters that a player can play competently in professional tournament matches.

2. Data Sources and Collection

There are two data sources used in the project. The primary dataset is the public match export from Oracle's Elixir, which includes data at the match level including tournament, date, player, team, role, selected champion, kills,

deaths, assists, gold, damage, and match result. The enrichment dataset is Riot champion metadata, which is stored in a simplified JSON file based on Riot's champion information, and is used to attach a wide range of tags for champion style.

The gathered sample is from 1 January 2024 to 14 September 2024 and includes the following events: LCK, LPL, LEC, Mid-Season Invitational, Esports World Cup, and Demacia Cup. Since the project is focused on the mid-lane role and the six selected players (Faker, Chovy, ShowMaker, Caps, knight and Bdd) it is limited to those.

Dataset overview

Item	Value
Match-level sample	732 player-match rows
Machine learning sample	282 post-patch rows
Players	Faker, Chovy, ShowMaker, Caps, knight, and Bdd
Patch windows	14.5, 14.13, and 14.16
Competitions	LCK, LPL, LEC, Mid-Season Invitational, Esports World Cup, and Demacia Cup

3. Data Preparation and Feature Engineering

Match metadata and player stats were joined by game identifier. The resulting table was filtered to the 2024 mid-lane sample. Each match was then linked with the champion metadata to allow grouping of champions by play style (Mage, Marksman, Assassin, etc.).

Three key patch transitions in 2024 were analysed. 28-day pre-patch and post-patch windows were set up for each patch. This enabled the project to make a comparison of the average player's behaviour both before and after the major update of the game.

A variety of match-derived metrics were also developed such as KDA, creep score per minute, damage per minute, gold per minute and kill participation. A composite performance measure was also created using six performance measures (win, KDA, creep score per minute, damage per minute, gold per minute, and kill participation) that were standardized to a z-score and then averaged.

In the patch-level summaries, the project calculated pre-patch champion-pool diversity, percentage of truly new champions in the post-patch window, and absolute difference in performance between pre-patch and post-patch windows. The hypothesis tests were conducted on these features, as well as the subsequent machine learning tasks.

4. Data Analysis: Exploratory Analysis and Hypothesis Testing

The first phase of the analysis compared the stability of Faker's performance after patch transitions with that of his peers, and tested the link between this potential stability and diversity in the champion pool. For the exploratory step, rolling visualizations, patch-level summary plots, and comparative tables were utilized. Formal inference was afterwards added by using permutation tests and Spearman correlation.

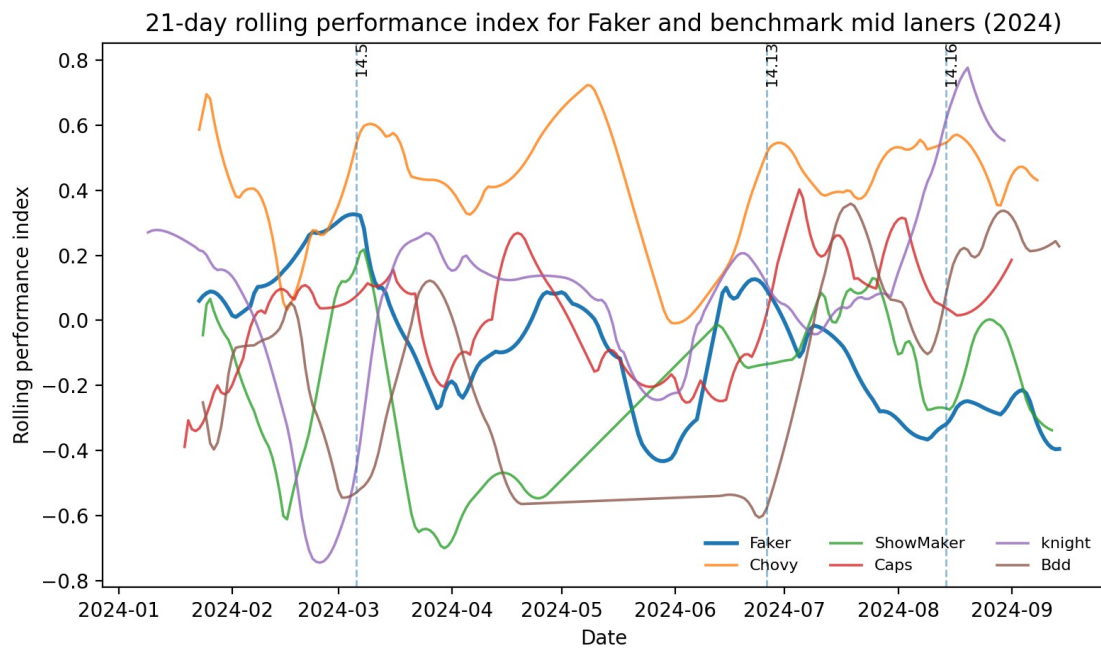


Figure 1. Rolling performance index during the 2024 season. Faker remains competitive across the year, but all players show visible variation around patch dates.

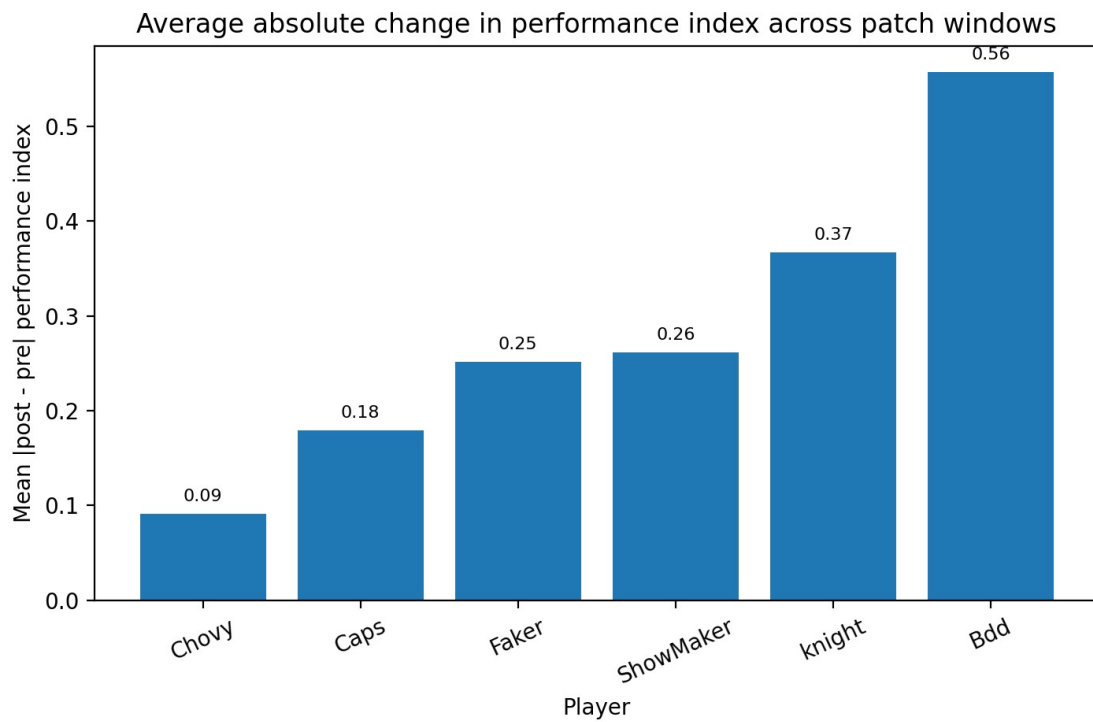


Figure 2. Mean absolute change in performance before versus after the three patch windows. Smaller values indicate greater stability.

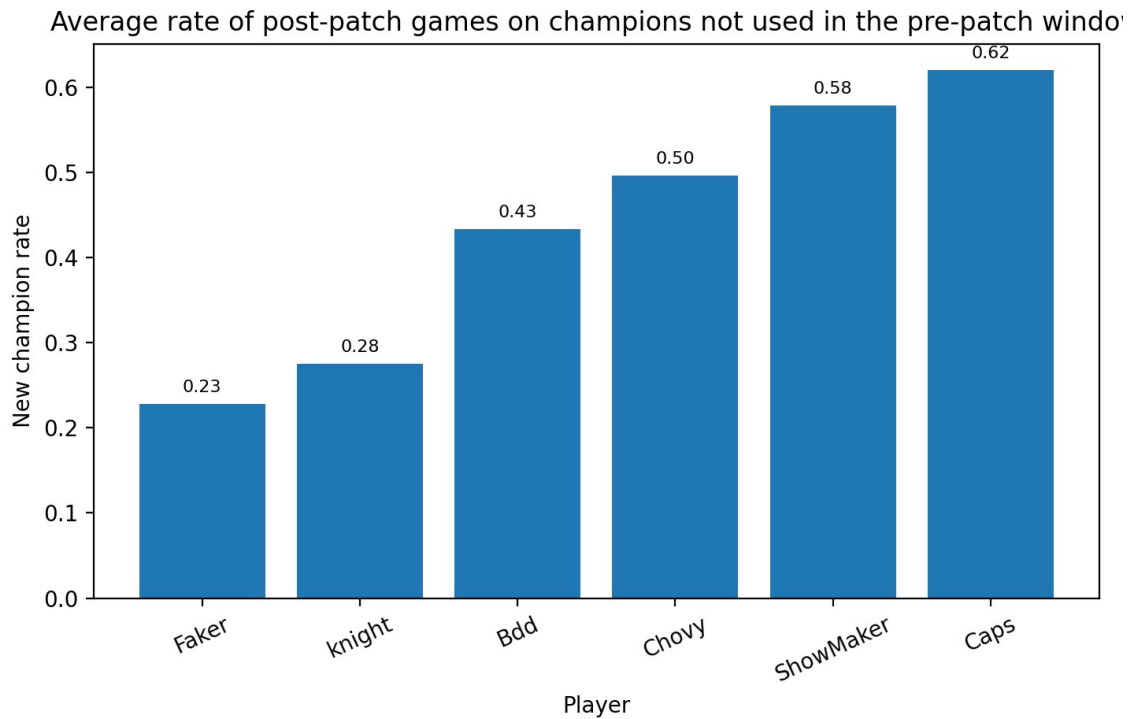


Figure 3. Mean rate of truly new champions used immediately after each patch. Faker appears more conservative than peers in this measure.

In descriptive terms, Faker's change in performance from pre- to post-patch was an average of 0.251, whereas his peers averaged 0.291. He had an average champion-pool diversity entropy of 2.399, while his peers had a diversity entropy of 2.197. The most descriptive pattern was for the use of truly new champions; Faker's mean usage of truly new champions was 0.228, while the peers were 0.481.

Hypothesis test results

Hypothesis	Method	Key value	Result
H1: Faker has smaller mean absolute post-patch performance change	Permutation test	$p = 0.436$	Directional support, not significant
H2: Faker has higher pre-patch champion-pool diversity	Permutation test	$p = 0.285$	Directional support, not significant
H3: Greater diversity is linked to smaller post-patch change	Spearman correlation	$\rho = -0.038$, $p = 0.880$	No meaningful relationship
H4: Faker uses fewer truly new champions after patches	Permutation test	$p = 0.063$	Strongest directional evidence, still not significant

As a whole, the first-stage analysis does not provide a compelling argument but rather a cautious interpretation. The direction of the observed curve suggests a slightly more conservative and slightly more stable post-patch curve for Faker, but the small sample size is not statistically compelling.

5. Data Analysis: Machine Learning

Essentially, this is all about dealing with data, specifically using machine learning techniques.

The second phase of the project was a process of turning the descriptive question into a predictive one. 282 rows of the post-patch machine learning dataset were generated. Pre-match information was used only in the context of the match to ensure the models simulate a plausible pre-match prediction situation.

A binary classification problem was set up. The first task asked if a match would be "stable" after the patch was applied (had the player been within 1 pre-patch standard deviation of their pre-patch overall). The second task determined whether a post-patch match would be high performance (performance index $\geq$ the post-patch sample median).

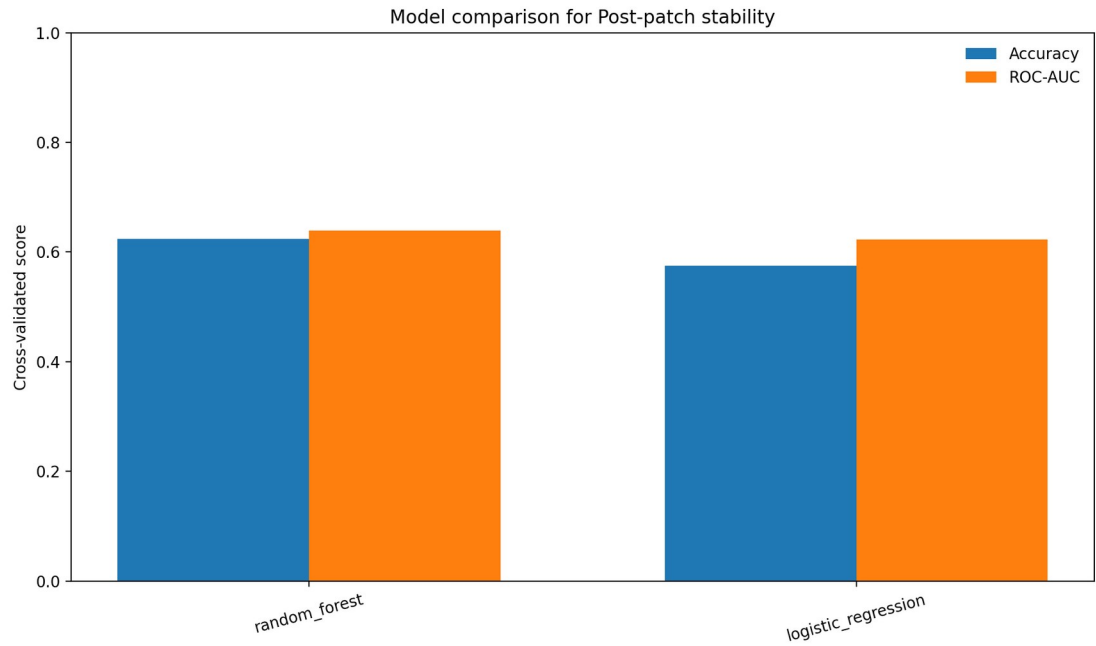


Figure 4. Cross-validated comparison of models for the post-patch stability task.

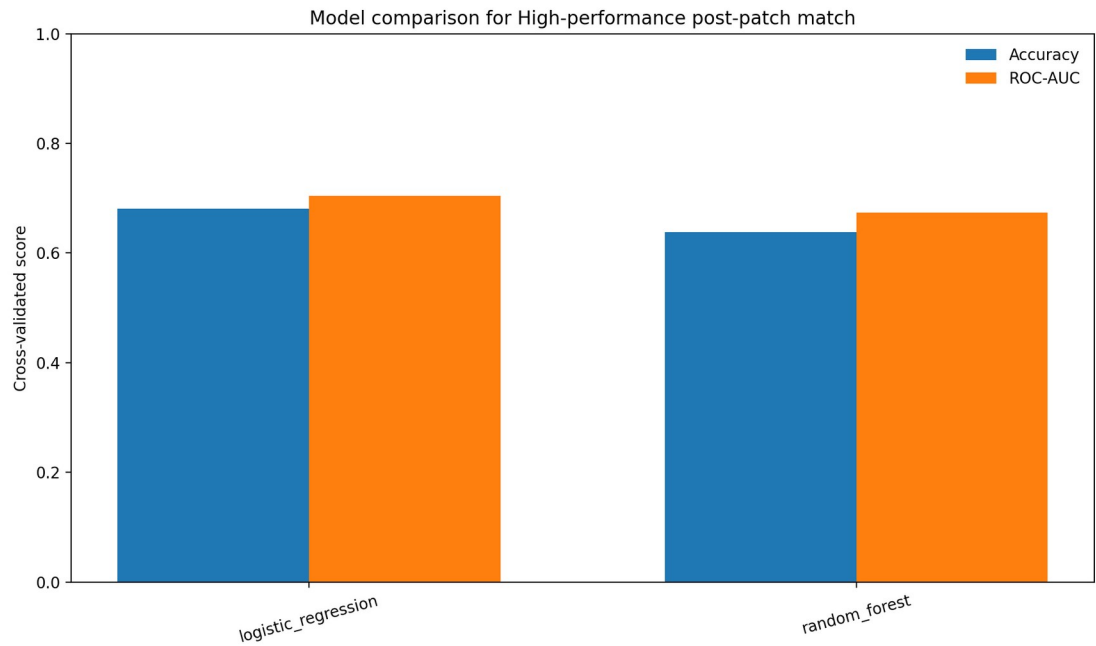


Figure 5. Cross-validated comparison of models for the high-performance post-patch task.

Cross-validated model comparison

Task	Best model	Accuracy	Balanced acc.	F1	ROC-AUC
Post-patch stability	random forest	0.624	0.612	0.679	0.639
High-performance post-patch match	logistic regression	0.681	0.681	0.668	0.704

Holdout evaluation on the final patch

There were features for player info, team info, league info, patch info, champion info, and champion type. Then there were a number of analytics features for averages in the pre-patch window, familiarity features for the champions and a number of rolling, recent-history, features from previous matches. Logistic regression and random forest were used and compared using 5-fold stratified cross-validation.

Task	Model	Accuracy	Balanced acc.	F1	ROC-AUC
Post-patch stability	random forest	0.612	0.529	0.727	0.558
High-performance post-patch match	logistic regression	0.506	0.480	0.618	0.537

The holdout results were significantly poorer compared to the cross-validated results, particularly for the task of high performance. The difference between the two is significant: there is more difficulty predicting behavior in a future patch than there is in randomly mixed folds. Collecting this limitation is not an issue with the project itself; rather, it is a useful result that provides knowledge about the challenges of patch-to-patch generalization.

6. Findings

- The mean absolute performance change (MAPC) value showed that Faker was more stable after the aforementioned patches in comparison with peers, with a value of 0.251 versus 0.291.
- There was also some evidence that Faker performed slightly better in patch windows in terms of pre-patch diversity entropy (2.399 compared to 2.197 for everyone else).
- The greatest descriptive difference was for post-patch conservatism, with fewer truly new champions being used by Faker immediately after patches (0.228 vs. 0.481).
- None of the hypothesis tests at this level of the process crossed the typical p-level 0.05 so there is not enough evidence to prove it.
- Moderate predictive performance was obtained with machine learning models. Stability had a cross-validated ROC-AUC value of 0.639 and high-performance matches had a cross-validated ROC-AUC value of 0.704.
- The holdout assessment of the last patch revealed that prediction of a future patch is not straightforward even when there is a strong historical signal of predictability.

7. Limitations and Future Work

There are certain restrictions of the project. First, it is a very specific sample, on six elite players, one role and three prominent patch windows in 2024. This reduces statistical power, but helps interpretability.

Secondly, the project relies on match-level player data and simplified champion data, rather than full draft context and opponent data, or richer team-level data. This does not capture as robustly as you might want how the rest of the league drafted, what the opposing team decided to pick, and how the player fits in with the rest of the team; all of this of course can affect a player's adaptation.

Thirdly the machine learning part is based on relatively normal classification models. For a course project, this is suitable, although more sophisticated time-aware validation schemes, sequence models, and/or more sophisticated draft embeddings might yield better results and more closely capture the temporal aspects of esports data.

This study could be expanded to cover additional seasons, patch windows, players and roles, and greater draft context. Comparing a player's adaptation profile to other players' profiles is straightforward, but comparing the adaptation profile to a team outcome such as an expected win probability based on post-patch champion selections would be another extension that is useful to make.

8. Reproducibility and Repository Contents

The course guidelines call for all the code to be written in Python. The data used is available both as raw and processed data, along with data analysis scripts, figures and Jupyter notebooks for both milestone 1 and milestone 2. To reproduce the analysis, install dependencies listed in requirements.txt and run the scripts located in the scripts folder of the repository for performing the milestone analysis and the machine learning.

Files of particular importance to the repository include: scripts/run_milestone_analysis.py, scripts/run_ml_analysis.py, a Jupyter notebook called milestone1_eda_hypothesis_tests.ipynb, a notebook called milestone2_ml_analysis.ipynb, the processed CSV results, and the final figures used throughout this report.

9. Conclusion

This project employed a Faker-centred data set for the game of League of Legends to tackle a larger question in data science: How resilient is the elite performer in the face of changing environments? The response of the existing sample is guarded yet intriguing. Faker does sound a bit more steady and more conservative than his peers following major patches, but the evidence is not showing it sufficiently to suggest a statistical certainty of the pattern.

Meanwhile, the machine learning findings reveal that certain post-patch behaviors can be anticipated by pre-match context and recent history of behavior. It is not only a descriptive sports analytics study that accomplished the project successfully, it also serves as a complete sports data science analysis pipeline - data collection, data enrichment, exploratory data analysis, hypothesis testing, predictive modelling, interpretation.